Ei Ei Nyein (EiEi) Chan
(361) 305- 8368 | eieinyeinchan10@outlook.com | LinkedIn
Edinburg, Texas, USA

PROFILE SUMMARY

Master's student in Computer Science at the University of Texas Rio Grande Valley with research experience in federated learning, trustworthy AI, and distributed machine learning systems. Current research focuses on personalized federated learning under non-IID settings and adaptive optimization. Presented research at TACCSTER 2025 (UT Austin) and received the Best Oral Presentation Award at UTRGV STEM Conference 2026. Research interests include federated learning, AI Systems, and parallel & distributed computing.

EDUCATION

Master of Science (Computer Science) — Expected: Dec 2026
University of Texas Rio Grande Valley — GPA: 3.82

Bachelor of Engineering (Computer Engineering) — Feb 2019
University of Technology, Yatanarpon Cyber City, Myanmar — GPA: 4.99/5.00

RESEARCH INTERESTS

Federated Learning
Trustworthy AI
AI Systems & Optimization
Distributed Machine Learning
Parallel & Distributed Systems

RESEARCH EXPERIENCE

Research Assistant (Incoming) | Security & Trustworthy AI Lab
Jun 2026 - Present

- Conduct research in trustworthy and personalized federated learning systems
- Investigate adaptive optimization and personalization under non-IID data settings
- Support manuscript preparation and experimental evaluation
- Mentor undergraduate students in AI/ML research activities

PUBLICATIONS & MANUSCRIPTS

- *Bayesian Optimization of PID Coefficients for Robust Federated Learning* (Manuscript in preparation)
- *A Survey: A Granularity-Centered Taxonomy of Personalized Federated Learning* (Master's thesis extension; manuscript in preparation, target submission 2026)

PRESENTATIONS & AWARDS

STEM Conference 2026 Oral Presentation, UTRGV (Best Oral Presentation Award, Graduate Level)

- Presented: "A Granularity-Centered Taxonomy of Personalized Federated Learning"

TACCSTER 2025 Poster Presentation, UT Austin

- Presented "Bayesian Optimization of PID Coefficients for Robust Federated Learning", using Python and HPC resources for scalable experimentation.

TECHNICAL SKILLS

Programming: Python, C++, Java, CUDA
Computing Platforms: Linux, HPC cluster (slurm), GPUs
Tools: Git, Jupyter, LaTeX

SELECTED RESEARCH PROJECTS

Parallel Computing project (Spring 2026)

- Implemented and evaluated Ring, Tree, and Butterfly AllReduce topologies using Python and CUDA for parallel/distributed computation.
- Analyzed communication efficiency and scalability in distributed environments.

Computer Architecture Project – GPU Performance Analysis (Fall 2025)

- Proposed and evaluated synchronous vs asynchronous GPU pipelining strategies inspired by Cypress task-based scheduling for tensor computation.
- Evaluated performance on NVIDIA A6000 and Tesla T4 GPUs, analyzing execution efficiency and speedup under varying matrix sizes and batching conditions.

AI Topics in Image Processing – MNIST Efficiency (Spring 2026)

- Proposed and developed an efficiency-focused MNIST study to analyze model performance and computational tradeoffs in image processing workflows.

Study on Hash Functions: SHA 512 and Whirlpool (B.Eng Final Year Thesis, UTYCC)

- Built a Python GUI application to compare SHA-512 and Whirlpool hash functions for secure data processing.

PROFESSIONAL EXPERIENCE

Reporting/Data Analyst, Accenture Malaysia, Kuala Lumpur (2021 – 2025)

- Performed data analysis using Python and Power BI to generate analytical reports supporting operational decision-making

LEADERSHIP & SERVICE

- Lab Member | Security & Trustworthy AI at UTRGV, Texas | June 2025 - Present
- Student Mentor & Peer Guide at UTYCC, Myanmar | 2015 – 2018

CERTIFICATIONS

- NVIDIA Fundamentals of Deep Learning
- Microsoft Power BI Analyst Professional Certificate

LANGUAGES

Burmese | English | Mandarin Chinese